

EMILIA PROTOCOL

The seatbelt for the agentic era.

Verifiable human authorization for the irreversible actions AI agents are starting to take — an open standard, verifiable offline by anyone.

Fifty years of security asked "who is allowed i

The dominant users of software are no longer humans — they're AI agents that move money, change records, and delete data on the fly. The unanswered question is no longer authentication. It's authorization at the moment of action:

Who approved **that exact action** — before it executed?

"A human was in the loop" is unfalsifiable.

- CISOs are **blocking agent deployment** — sitting on AI budget they can't spend because compliance can't answer "who authorized that?"
- Regulators now **require** human oversight (EU AI Act Art. 14) — but no one can **verify** the fact.
- Decision logs are **testimony** — what the operator says happened, in a database the operator controls.

Authorization receipts.

Before a high-risk action runs, a **named human** approves the *exact* action on their own device (WebAuthn / Face ID). EP emits a signed evidence packet anyone can verify **offline, with no account, without trusting our servers.**

Alter one byte and verification fails. The receipt stays valid even if EMILIA disappears. **`npx @emilia-protocol/crash-test`**

Not a pitch. A protocol.

3 langs · 8 suites

JavaScript / Python / Go verifiers agree on 8 adversarial cross-language conformance suites — the IETF bar for a standard.

Machine-checked

TLA+ safety properties + Alloy relational models, verified in CI on every change.

Independently verified

An outside implementer ran the crash-test + conformance on their own machine and confirmed (2026). First external reliance event.

IETF drafts

Authorization receipts + multi-party quorum protocol. Internet-Drafts; evidence-chain forthcoming.

We didn't enter a crowded field. We became its center.

A dozen-plus efforts now define "a receipt for an agent's action" — delegation, policy-permit, decision, compliance. They all converged on the same substrate. **None composes them, and only EP binds an accountable human.**

EP authored the **Authorization Evidence Chain** — the thin layer that verifies the whole chain for one action, offline. EP is the convergence point, and the only one with the human leg.

Healthcare.

EP's two-person rule is **already mandated practice**: the ISMP / Joint Commission *independent double-check* for high-alert medications. EP turns that attestation into tamper-evident, offline-verifiable evidence — **PHI-free by construction**.

Same primitive covers capital procurement (kills payment-redirect fraud) and answers Article 14 for medical AI. Go-to-market: healthcare-AI vendors + procurement platforms embed EP (B2B2H), not direct hospital sales.

Where high-risk actions live.

Government

Fraud, benefit routing, operator overrides. A \$500M state fraud finding engaged.

Healthcare

High-alert med double-check, capital procurement, Article 14 medical AI.

Finance

Payment-change / beneficiary fraud, treasury disbursements, privileged approvals.

Agents

Execution controls for autonomous and agents across all of the above.

The standard is free. The trust root is the business.

- **Free & permissionless:** the format and verifier — that creates ubiquity.
- **Revenue layer:** the operated trust root — enrollment / approver directory, federated transparency log, revocation, compliance-grade evidence pipelines — that an organization cannot casually self-host.
- **Plus:** hosted verification, sector policy packs, audit & evidence tooling, enterprise support.

Precedent → recommendation → requirement

Public-key crypto existed two decades before the browser padlock made its *absence* unacceptable. EP becomes obvious the same way: when a party with leverage makes the **absence of a receipt** unacceptable —

- an insurer's coverage condition on wire / benefit fraud,
- an auditor's controls framework,
- a government AI-auditor registry that needs something verifiable to audit against.

"The required human approvals existed for this exact action before it executed — and I verified that myself, without trusting the vendor, the operator, or any log."

EMILIA Protocol · open standard, IETF-track, independently verified ·
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